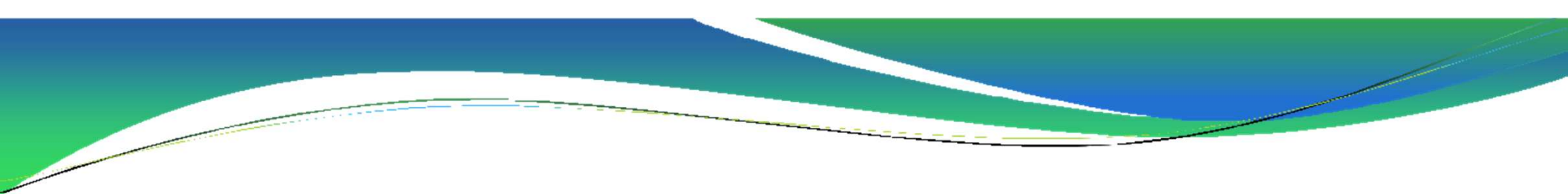
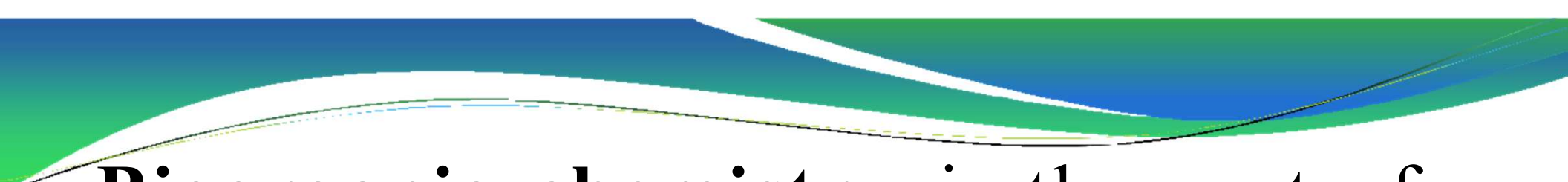


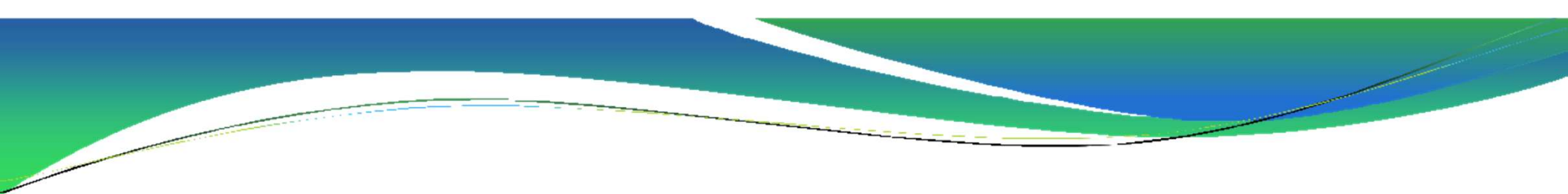
Classification and the nomenclature of organic compounds.



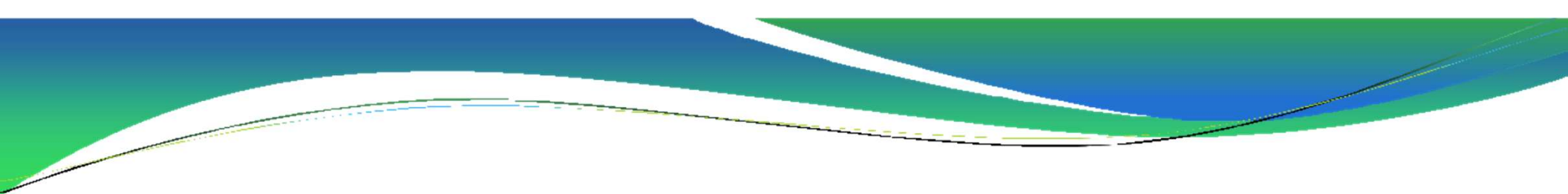
Organic chemistry is the
chemistry of compounds of
carbon.



Bioorganic chemistry is the part of organic chemistry that studies the carbon compounds, which are present in the living organism – the so-called **biomolecules**. The major biomolecules are carbohydrates, proteins, lipids, and nucleic acids.



Bioorganic chemistry also studies organic chemicals that are applied in medicine as pharmaceutical drugs.

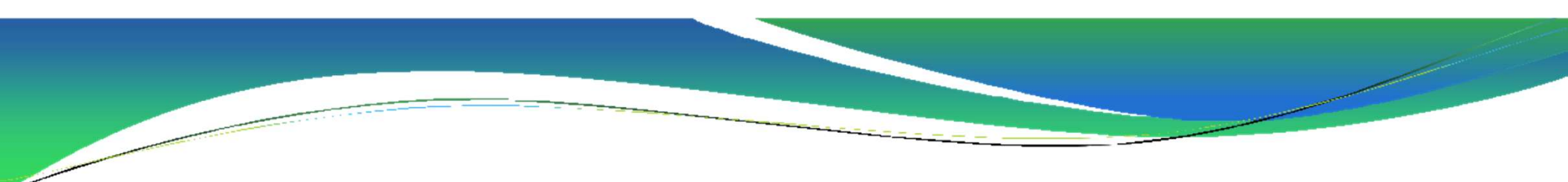


Hydrocarbons are organic chemicals that contains only carbon and hydrogen.

Saturated hydrocarbons are compounds that contains only single bonds between carbon atoms.

Alkanes contain only single covalent bonds between carbon atoms and have the general formula C_nH_{2n+2}

Unsaturated hydrocarbons are hydrocarbons that contains one or more double or triple bonds.



Alkenes are hydrocarbons that contains a double bond.

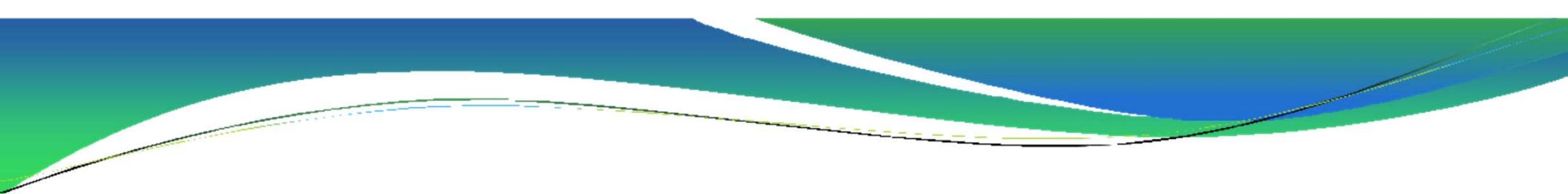
The general formula is C_nH_{2n} .

Alkynes are hydrocarbons that contains a triple bond.

The general formula is C_nH_{2n-2} .

Cyclic compounds are compounds that contains a ring (cycle).

Carbocyclic compounds are compounds that include chains of carbon atoms closed in a ring.



Alicyclic (aliphatic cycle) are nonaromatic cycle organic compounds.

Aromatic hydrocarbons have carbon atoms arranged in hexagonal rings, based on benzene C_6H_6 .

Heterocyclic compounds are compounds that have certain heteroatoms (predominantly N, O or S) included in the structure of their cycles.

Organic compounds

Open chain or Aliphatic

Closed chain or Cyclic

Saturated hydrocarbons

Unsaturated hydrocarbons

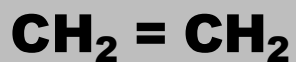
Carbocyclic or Homocyclic

Heterocyclic

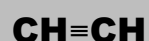
Alkanes



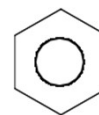
Alkenes



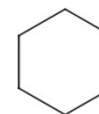
Alkynes



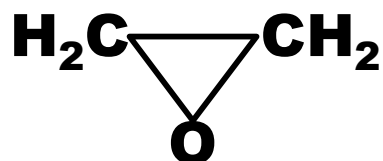
Aromatic (arenes)



Alicyclic

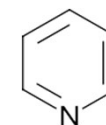


Hetero alicyclic

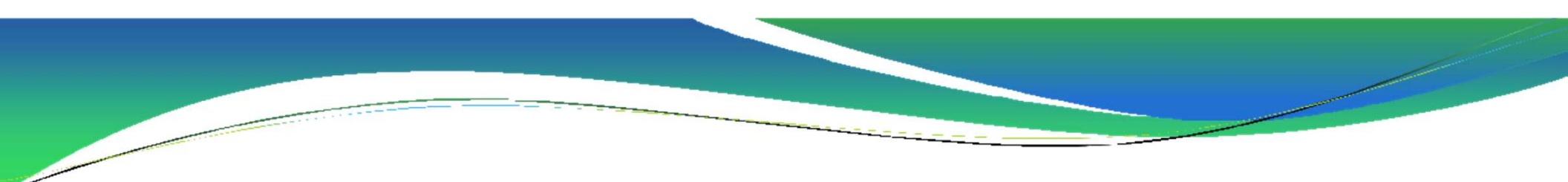


Ethelene oxide

Hetero aromatic



Pyridine

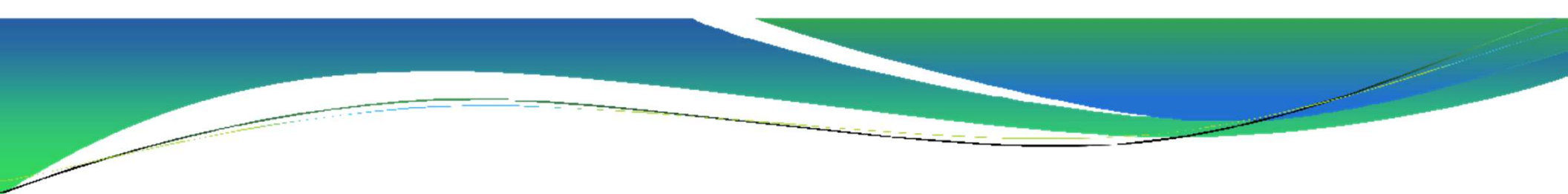


Acyclic compounds (or aliphatic) are organic chemicals having open chains of carbon atoms, branched or unbranched. For example:

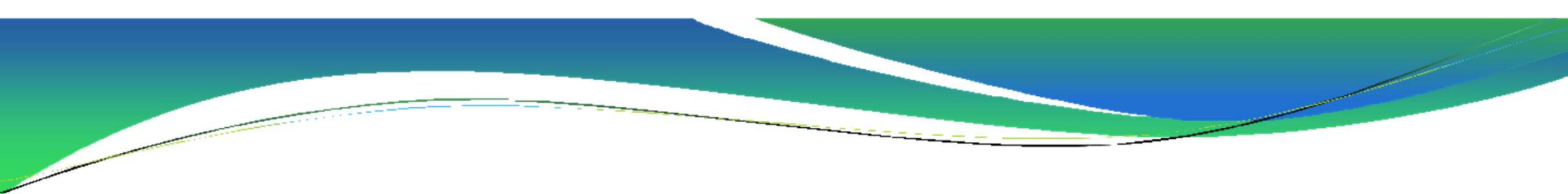
$\text{CH}_3 - \text{CH}_2 - \text{CH}_3$ unbranched
propane

$$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{CH} - \text{CH}_3 \end{array}$$
 branched

2 – methylpropane



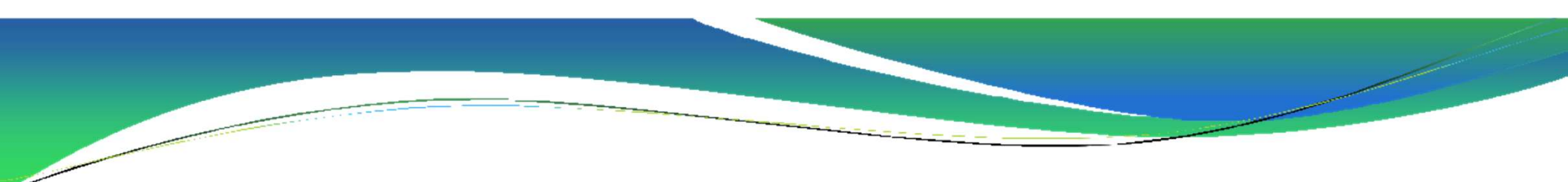
Functional groups



A **functional group** is an atom or group of atoms attached to a hydrocarbon residue, R. The functional group often confers specific properties to an organic molecule.

NAME OF CLASS	GENERAL FORMULA	FUNCTIONAL GROUP		IUPAC	
		FORMULA	NAME	PREFIX	SUFFIX
<i>Carboxylic acids</i>	<i>RCOOH</i>	<i>-(C)OOH[*]</i>	<i>Carboxyl</i>	-----	<i>-oic acid</i>
		<i>-COOH</i>		<i>Carboxy-</i>	
<i>Sulfonic acids</i>	<i>R-SO₃H</i>	<i>-SO₃H</i>	<i>Sulfonate</i>	<i>Sulfo-</i>	<i>- sulfo acid</i>
<i>Esters</i>	<i>R-COO-R</i>	<i>-(C)OOR[*]</i>	<i>Carbalkoxy</i>	-----	<i>-oate</i>
<i>Amides</i>	<i>R-CONH₂</i>	<i>-CONH₂</i>	<i>Amide</i>	<i>Carbamyl-</i>	<i>-amide</i>
<i>Nitriles</i>	<i>R-C≡N</i>	<i>-(C)≡N[*]</i>	<i>Nitrile</i>	-----	<i>-nitrile</i>
		<i>-CN</i>		<i>Cyano-</i>	-----

<i>Aldehydes</i>	$R-\overset{\overset{O}{\parallel}}{C}-H$	$-(C)\overset{\overset{O}{\parallel}}{H}$	<i>Carbonyl</i>	<i>Formyl-</i>	<i>-al</i>
<i>Ketones</i>	$R-CO-R$	$>C=O$		<i>Oxo or keto-</i>	<i>-one</i>
<i>Alcohols</i>	$R-OH$	$-OH$	<i>Hydroxyl</i>	<i>Hydroxy-</i>	<i>-ol</i>
<i>Thioalcohols</i>	$R-SH$	$-SH$	<i>Thiol</i>	<i>Mercapto-</i>	<i>-thiol</i>
<i>Amines</i>	$R-NH_2$	$-NH_2$	<i>Amino</i>	<i>Amino-</i>	<i>-amine</i>
<i>Nitro compound</i>	$R-NO_2$	$-NO_2$	<i>Nitro</i>	<i>Nitro-</i>	-----
<i>Ethers</i>	$R-O-R$	$-OR$	<i>Alkoxy</i>	<i>Alkoxy-</i>	-----
<i>Halogen derivatives</i>	$R-Hal$	$-Hal(-F, -Cl, -Br, -I)$	<i>Halogens</i>	<i>Fluoro-, chloro-, bromo-, iodo-</i>	-----



Nomenclature of organic compound (IUPAC)



Nomenclature

1. Identify the Longest Carbon Chain (Parent Chain)

- Select the longest continuous chain of carbon atoms as the parent chain.
- This determines the base name of the compound.

Example:

For $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$

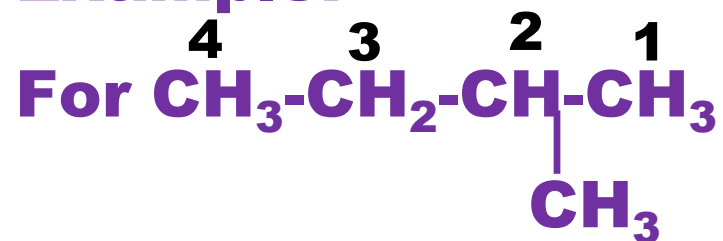
the parent chain has 4 carbons → Butane.

Nomenclature

2. Number the Carbon Atoms in the Parent Chain

- Number the chain starting from the end nearest to a substituent group.
- If there is a tie, prioritize functional groups or double/triple bonds.

Example:



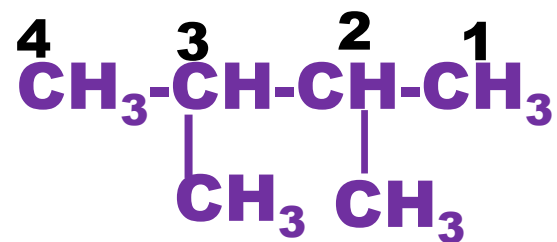
numbering starts from the right so the substituent is on carbon 2 → 2-Methylbutane.

Nomenclature

3. Identify and Name Substituents

- **Substituents are groups attached to the parent chain, such as alkyl groups or halogens or any others.**
- **Name substituents alphabetically, and use prefixes like di-, tri-, tetra- for multiple identical groups.**

Example:



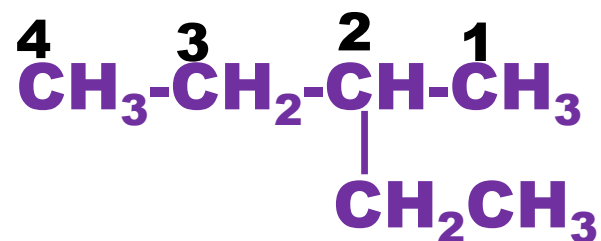
the substituents are methyl groups → 2,3 dimethylbutane.

Nomenclature

4. Assign Locants to Substituents

- Assign numbers to substituents based on their position on the parent chain.
- Use the smallest possible set of numbers.

Example:

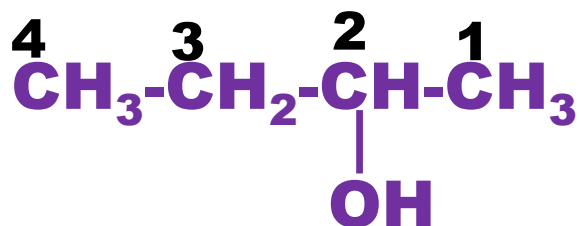


the substituent ethyl is on carbon 2 → 2-Ethylbutane.

Nomenclature

5. Incorporate Functional Groups

- **Functional groups have specific suffixes or prefixes.**
- **Functional groups are prioritized over alkyl and halogen substituents for numbering.**



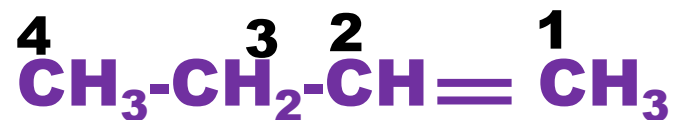
the suffix for the alcohol group is -ol → Butan-1-ol.

Nomenclature

6. Indicate Multiple Bonds

- Double bonds use -ene, and triple bonds use -yne.
- Use prefixes like di-, tri-, etc., for multiple double or triple bonds.

Example:



the double bond is at position 1 → But-1-ene.

Nomenclature

7. Combine the Name

- Write the substituents, parent chain, and functional group together, following alphabetical order and numerical priority.

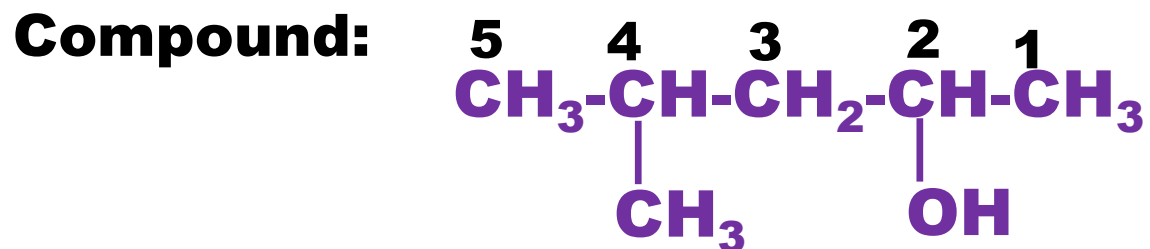
8. Cycloalkanes and Aromatic Compounds

- For cyclic compounds, use the prefix **cyclo-**.
- For aromatic compounds, the base name is often **benzene**.

Example: For a 6-carbon ring with a double bond and a methyl group → Methylcyclohexene.

Nomenclature

Example of Full Name Combination



- **Parent chain: Pentane**
- **Substituents: Methyl at position 4 and Hydroxyl at position 2**
- **Final Name: 2-Hydroxy-4-methylpentane.**



Priority list of functional group in Nomenclature

1.Carboxylic acid

2.Sulfonic acid

3.Esters

4.Acid halides

5.Amides

6.Cyanides

7.Aldehydes

8.Ketones

9.Alcohols

10.Amines

11.Alkynes

12.Alkenes

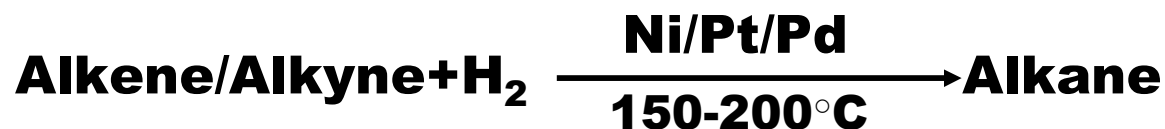
13.Alkanes

Preparation of alkane

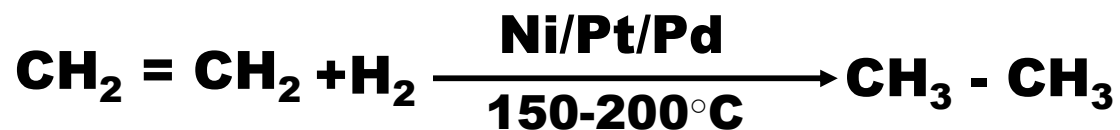
1. From Unsaturated Hydrocarbons (Hydrogenation)

Reaction: Addition of hydrogen (H_2) to alkenes or alkynes in the presence of a catalyst like nickel (Ni), platinum (Pt), or palladium (Pd) at $150-200^\circ C$

▪



Example:



2. Wurtz Reaction

Reaction: Alkyl halides are treated with sodium (Na) in dry ether to form alkanes with an even number of carbon atoms.



Example:



(Methyl bromide to ethane)

3. Reduction of Alkyl Halides

Reaction: Alkyl halides are reduced to alkanes using zinc (Zn) and an acid (e.g., HCl).



Example:



(Methyl chloride to methane)

4. Decarboxylation of Carboxylic Acids

Reaction: Carboxylic acids or their sodium salts react with soda lime (NaOH + CaO) to produce alkanes.



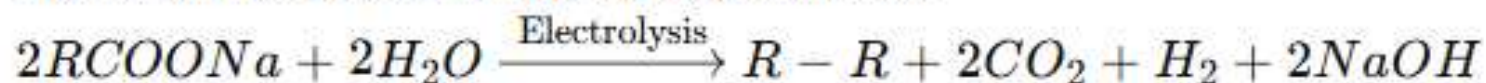
Example:



(Sodium acetate to methane)

5. Kolbe's Electrolysis

Reaction: Electrolysis of aqueous solutions of sodium or potassium salts of carboxylic acids produces alkanes with an even number of carbon atoms.



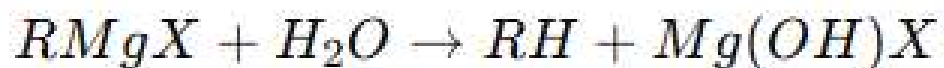
Example:



(Sodium acetate to ethane)

6. Grignard Reagent Reaction

Reaction: Alkanes can be prepared by treating Grignard reagents (RMgX) with water or alcohol.



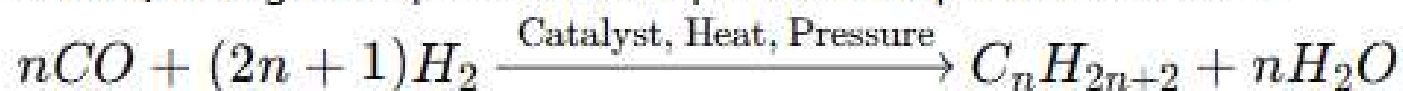
Example:



(Methyl magnesium bromide to methane)

7. From Carbon Monoxide and Hydrogen (Fischer-Tropsch Process)

Reaction: A mixture of carbon monoxide (CO) and hydrogen (H₂) is passed over a catalyst (iron or cobalt) at high temperature and pressure to produce alkanes.



Example:

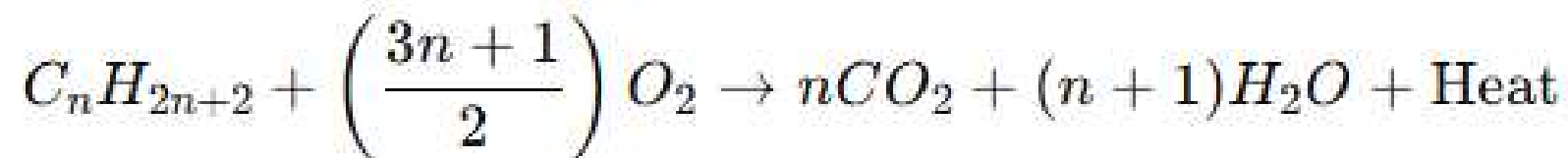


Reaction of Alkane

1. Combustion

Alkanes react with oxygen to produce carbon dioxide, water, and heat.

General Reaction:



Example: Combustion of methane:



2. Halogenation (Substitution Reaction)

Alkanes react with halogens (Cl_2 , Br_2) in the presence of sunlight or UV light, leading to the substitution of hydrogen atoms with halogen atoms.

General Reaction:



Example: Chlorination of methane:



Subsequent halogenation can occur to form more substituted products:



3. Cracking (Thermal or Catalytic)

Alkanes can be broken into smaller alkanes and alkenes at high temperatures or in the presence of catalysts.

General Reaction:



Example: Cracking of decane:



4. Isomerization

Straight-chain alkanes can be converted into branched-chain alkanes in the presence of a catalyst (e.g., $\text{AlCl}_3 + \text{HCl}$).

Example:



5. Aromatization (Reforming)

Alkanes with six or more carbon atoms can be converted into aromatic compounds under high temperatures and pressure, using catalysts such as chromium, molybdenum, or platinum.

Example:



(Hexane to benzene)

6. Pyrolysis

Alkanes decompose into smaller hydrocarbons when heated strongly in the absence of air.

Example:



Preparation of alkene

1. Dehydration of Alcohols

Reaction: Alcohols undergo dehydration (removal of water) in the presence of concentrated sulfuric acid (H_2SO_4), phosphoric acid (H_3PO_4), or alumina (Al_2O_3) as a catalyst to form alkenes.



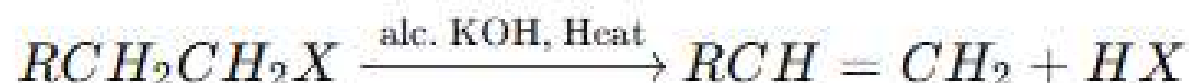
Example:



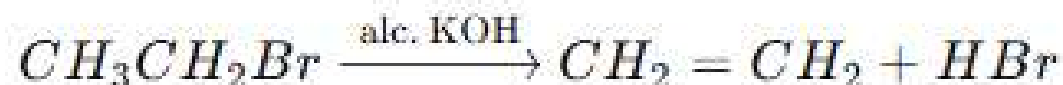
(Ethanol to ethene)

2. Dehydrohalogenation of Alkyl Halides

Reaction: Alkyl halides lose a hydrogen atom and a halogen atom (dehydrohalogenation) in the presence of alcoholic potassium hydroxide (*alc.KOH*) to form alkenes.



Example:



(Ethyl bromide to ethene)

3. Cracking of Alkanes

Reaction: Long-chain alkanes are broken down into shorter-chain alkanes and alkenes by heating them in the presence of a catalyst.



Example:



(Hexane to ethene and butane)

4. Elimination from Vicinal Dihalides

Reaction: Vicinal dihalides (compounds with halogens on adjacent carbons) undergo dehalogenation in the presence of zinc to form alkenes.



Example:



(1,2-dibromoethane to ethene)

5. Wittig Reaction

Reaction: Alkenes are formed when aldehydes or ketones react with Wittig reagents (phosphonium ylides).



Example:



(Acetaldehyde to ethene)

6. Kolbe's Electrolysis

Reaction: Electrolysis of aqueous solutions of sodium or potassium salts of carboxylic acids produces alkenes.



Example:



(Sodium acetate to ethene)

7. Partial Hydrogenation of Alkynes

Reaction: Alkynes undergo partial hydrogenation to form alkenes in the presence of catalysts like Lindlar's catalyst (Pd/CaCO₃ poisoned with quinoline).



Example:



(Ethyne to ethene)

8. From Grignard Reagents

Reaction: Grignard reagents react with alkyl halides to form alkenes.



Example:



(Methyl magnesium bromide to ethene)

Reaction of alkene

1. Combustion

Alkenes burn in oxygen to produce carbon dioxide and water, releasing heat.

General Reaction:



Example:



(Ethene combustion)

2. Addition Reactions

Alkenes readily undergo addition reactions across the double bond.

(a) Hydrogenation (Addition of H_2)

Alkenes react with hydrogen in the presence of a catalyst like Ni, Pt, or Pd to form alkanes.



Example:



(Ethene to ethane)

(b) Halogenation (Addition of X_2)

Alkenes react with halogens (e.g., Cl_2 , Br_2) to form dihaloalkanes.



Example:



(Ethene with bromine produces 1,2-dibromoethane)

(c) Hydrohalogenation (Addition of HX)

Alkenes react with hydrogen halides (e.g., HCl, HBr) to form haloalkanes. Markovnikov's rule applies: the hydrogen atom bonds to the carbon with more hydrogen atoms.



Example:



(Ethene with HBr forms ethyl bromide)

(d) **Hydration (Addition of H_2O)**

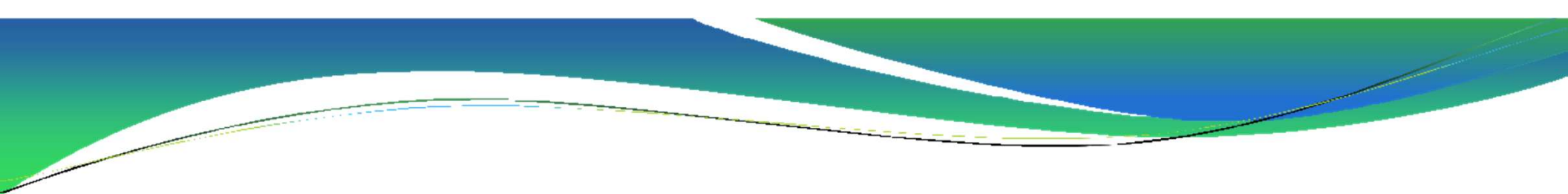
In the presence of acid catalysts (e.g., H_2SO_4), alkenes react with water to form alcohols.



Example:



(Ethene to ethanol)



(e) Oxidation (Epoxidation)

Alkenes react with peroxyacids (e.g., m-chloroperbenzoic acid, *MCPBA*) to form epoxides.



Example:



(Ethene to ethylene oxide)

3. Polymerization

Alkenes undergo polymerization to form polymers like polyethene.

Example:



(Ethene polymerizes to polyethene)

4. Oxidation Reactions

(a) Cold, Dilute KMnO_4 (Baeyer's Test)

Alkenes react with cold, dilute potassium permanganate to form diols. The purple color of KMnO_4 disappears.



Example:



(Ethene to ethylene glycol)

(b) Oxidation with Hot KMnO_4 or Ozonolysis

Alkenes oxidize with hot, concentrated KMnO_4 or ozone (O_3) to form carboxylic acids, ketones, or carbon dioxide.

Ozonolysis Example:



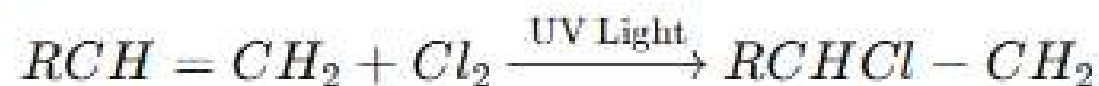
(Ethene to formaldehyde)

Hot KMnO_4 Example:



5. Allylic Substitution

In the presence of Cl_2 or Br_2 and UV light, alkenes undergo substitution at the allylic position.



Example:



(Ethene to ethylene dichloride)

Preparation of Alkyne

1. Dehydrohalogenation of Dihalides

Reaction: Vicinal or geminal dihalides are treated with alcoholic potassium hydroxide (*alc. KOH*) or sodium amide ($NaNH_2$) to remove two molecules of hydrogen halide (HX) and form an alkyne.

From Vicinal Dihalides:



Example:



(1,2-dibromoethane to ethyne)

From Geminal Dihalides:



Example:



(1,1-dibromoethane to ethyne)

2. Partial Dehydrogenation of Alkanes

Reaction: Alkanes are heated to high temperatures in the presence of catalysts to produce alkynes through partial dehydrogenation.



Example:



(Ethane to ethyne)

3. From Calcium Carbide (Industrial Method)

Reaction: Calcium carbide reacts with water to produce ethyne (acetylene).



Example:



(Calcium carbide to ethyne)

4. Kolbe's Electrolysis

Reaction: Electrolysis of aqueous solutions of sodium or potassium salts of dicarboxylic acids results in the formation of alkynes.



Example:



(Sodium oxalate to ethyne)

5. From Tetrahalides

Reaction: Tetrahalides undergo reductive elimination with zinc to form alkynes.



Example:



(1,1,2,2-tetrabromoethane to ethyne)

6. Alkylation of Acetylene

Reaction: Terminal alkynes ($HC \equiv CH$) react with alkyl halides in the presence of a strong base (e.g., sodium amide, $NaNH_2$) to form higher alkynes.



Example:



(Ethyne to propyne)

Reactions of Alkyne

1. Addition Reactions

Alkynes undergo electrophilic addition reactions similar to alkenes, but because of the triple bond, the reactions can happen in two steps (first breaking one pi bond and then the second).

a) Hydrogenation

- **Catalytic Hydrogenation:** Alkynes can be fully hydrogenated to alkanes in the presence of a metal catalyst like Pt, Pd, or Ni.



- **Partial Hydrogenation:** Using Lindlar's catalyst, alkynes are hydrogenated to *cis-alkenes* (stops at one pi bond).



b) Halogenation

- Alkynes react with halogens (Cl_2 , Br_2) to form dihalides or tetrahalides.
- Addition of 1 mole of halogen gives a dihalide (trans-alkene).



- Excess halogen gives a tetrahalide.



c) Hydrohalogenation

- Alkynes react with hydrogen halides (HCl , HBr) to form vinyl halides or dihalides, depending on the stoichiometry.



- Excess HX adds another molecule, leading to a geminal dihalide.



d) Hydration (Acid-Catalyzed)

- In the presence of dilute sulfuric acid and mercuric sulfate (HgSO_4), alkynes undergo hydration to form enols, which tautomerize to ketones.



e) Hydroboration-Oxidation

- Alkynes react with borane (BH_3) or bulky boranes like 9-BBN, followed by oxidation with hydrogen peroxide (H_2O_2) to yield aldehydes or ketones.



2. Oxidation Reactions

a) Ozonolysis

- Alkynes can be cleaved using ozone (O_3), followed by hydrolysis, to form carboxylic acids.



b) Oxidation with $KMnO_4$

- Potassium permanganate oxidizes alkynes to diketones under mild conditions.



3. Acidity of Terminal Alkynes

- Terminal alkynes ($RC \equiv CH$) have acidic hydrogen atoms. The proton can be abstracted by strong bases like sodium amide ($NaNH_2$) or organolithium reagents to form an alkynide ion.



